

DR.DEEPAK PATEL CHITYALA

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SUMMARY

A highly analytical and results-driven healthcare professional with over 3 years of experience in data analysis, research, and healthcare data management. Proven ability to leverage programming languages (**Python, R, SAS**) and data visualization tools (**Excel, Tableau, Power BI**) to derive actionable insights and improve data quality. Adept at using electronic medical records systems **e-ClinicalWorks, CERNER, EPIC**.

EDUCATION

Master of Science in Health Informatics

Indiana University – Purdue University

December 2023

Indianapolis, IN

Bachelor of Dental Surgery

DR. NTR University of Health Sciences

April 2021

Hyderabad, India

SKILLS

Languages: SQL, Python, R, SAS.

Databases: MySQL, SQL Server, MS Access, Web of Science.

Visualization: Excel, Tableau, Microsoft Power BI.

Electronic Medical Records: e-Clinical Works, CERNER, REDCap, EPIC Systems.

Other Skills: HL7 Interoperability Standards, Medical coding (CPT & ICD), Telemedicine, EHR Librehealth, Unified modeling language (UML), Medicare, Medicaid, HIPAA, Electronic case report forms (eCRFs), HIPAA, XML, HEDIS.

WORK EXPERIENCE

Clinical Data Analyst

Aug 2023- Present

Indiana University School of Medicine (IUSM)

Indianapolis, IN

- Investigated the role of T-follicular regulatory cells in food allergy and airway inflammation models, aiming to elucidate their impact on human diseases using animal models.
- Leveraged **Excel spreadsheets and SQL** to Execute, Transform, and Load (ETL) differential gene expression/RNA genomic sequence data.
- Utilized MySQL to efficiently manage, organize, and preprocess a dataset encompassing approximately 14,000 genes across different cells.
- Conducted data cleaning, transformation, and structuring via Excel and SQL, resulting in a 20% reduction in dataset errors.
- Employed data visualization tools to generate interactive graphical representations, revealing pivotal insights from extensive datasets.
- Utilized Python libraries including **Matplotlib, Pandas, and Pyplots** to construct Venn diagrams illustrating common genes shared in cells.
- Achieved a 95% accuracy rate in identifying common genes across three T-follicular cells (HDM, PCT, and Naive).

Research Assistant

Jan 2022- Jul 2023

Health Information Management (HIM), Indiana University

Indianapolis, IN

- Shouldered research projects encompassing literature reviews, data gathering, and analysis within health information management.
- Developed automated SQL scripts for routine data extraction and reporting, utilizing complex joins, subqueries, and window functions to improve efficiency and accuracy.
- Implemented scheduled data refreshes in Power BI, leveraging Direct Query and data modeling techniques without manual intervention.
- Created dynamic **Power BI reports with DAX measures**, calculated columns, and configured alerts for critical metric changes.
- Developed SQL scripts to automate data manipulation tasks, including data cleansing and transformation, streamlining workflows and enabling quicker access to insights.

Healthcare Data Analyst

Apr 2020- Nov 2021

Malla Reddy Super Specialty Hospitals

Hyderabad, India

- Collaborated with healthcare professionals and doctors to comprehensively grasp medical terminology, procedures, treatments, and EHR.
- Detected counterfeit insurance claims, resulting in a 15% reduction in financial losses.
- Enhanced claim processing efficiency by 20% by implementing streamlined procedures.
- Analyzed and interpreted clinical data to enhance patient care, inform decision-making, facilitate patient billing, and schedule appointments, resulting in a significant 40% enhancement in the quality of medical care processes.
- Acquainted with electronic health record (**EHR**) systems utilized in healthcare practices and achieved 30% reduction in patient record errors.

PROJECTS

Using AI-Enhanced Wearable Technologies For Real-Time Monitoring Multiple Sclerosis Patients

Spring 2023

Indiana University- Purdue University

Indianapolis, IN

Technologies used: Excel, Python (Jupyter hub), PubMed

- Leveraged Excel, Python (Jupyter Hub), and PubMed to integrate machine learning algorithms and artificial intelligence into wearable technologies to monitor Multiple Sclerosis patients continuously.
- Enabled real-time insights with 95% accuracy and personalized recommendations for managing individual symptom patterns through AI adaptation.

Prediction Of COVID-19 Using Blood Parameters

Spring 2022

Indiana University - Purdue University

Indianapolis, IN

Technologies used: SQL, Python (Jupyter hub), Excel spreadsheets, Kaggle

- Developed a Machine Learning model (Binary classification) leveraging SQL, Python (Jupyter Hub), Excel spreadsheets, and Kaggle to predict the relationship between blood indicators and their impact on COVID-19.
- Evaluated the correlation between blood parameters and COVID-19 using Python and SQL visualization techniques and heatmaps.
- Achieved a 90% accuracy rate in predicting positive COVID-19 results based on blood parameters, optimizing project efficiency.